Python Programming - 2301CS404

Lab - 8

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

In [12]:
```
n =  set(map(int,input("enter number :").split()))
m =  set(map(int,input("enter number for list 2 :").split()))
print(n.intersection(m))
print(n.union(m))
print(n-m)
```

```
{1, 2, 3}
{1, 2, 3, 4, 5, 6, 7, 8, 9}
{4, 5, 6}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

In [14]:
```
n =  set(map(int,input("enter number :").split()))
m =  set(map(int,input("enter number list 2 :").split()))
p =  set(map(int,input("enter number list 3 :").split()))
ans =  p.intersection(n.intersection(m))
print(ans)
```

```
{1, 2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

In [15]:
```
n =  list(map(int,input("enter number :").split()))
ans = set(n)
print(ans)
```

```
{1, 2, 3, 4, 5}
```

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

In [19]:
```
n =   set(map(int,input("enter number :").split()))
m =   set(map(int,input("enter number :").split()))
ans = n.intersection(m)
print('common elements between us :',len(ans))
```

```
common elements between us : 3
```

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

In [61]:
```
dict1 = {1: 10, 2: 20, 3: 30}
dict2 = {2: 40, 3: 50, 4: 60}
dict4 = {}
s = 0
print(type(dict1),dict1)
for i in dict1:
    for j in dict2:
        if(i==j):
              s =  dict1.get(i)+ dict2.get(j)
              dict4[i]=s
        else:
            dict4[i] = dict1.get(i)

print(dict4)
```

```
<class 'dict'> {1: 10, 2: 20, 3: 30}
{1: 10, 2: 20, 3: 30}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

In [90]:

```
l1 = list(input("Enter subjects (comma separated): ").split(","))
l2 = list(map(int, input("Enter marks (comma separated): ").split(",")))

d = dict(zip(l1, l2))
print("Original Dictionary:", d)

def sort_by_value(item):
    return item[1]

sorted_dict = dict(sorted(d.items(), key=sort_by_value))

print("Sorted Dictionary:", sorted_dict)
```

```
Original Dictionary: {'math': 70, 'physics': 85, 'chem': 60}
Sorted Dictionary: {'chem': 60, 'math': 70, 'physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

In [93]:

```
d1 = {'A': 50, 'B': 90, 'C': 30}

m = max(d1,key=d1.get)
n = min(d1,key=d1.get)
print('max is ', m)
print('min is ', n)
```

```
max is  B
min is  C
```

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

In [63]:
```
n = int(input("enter number"))
s = {}

for i in range(1,n+1):
    s[i] = i**2
print(s)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

In [73]:
```
w = input("enter word")
s1 = {}
count = 0
for i in w:
    for j in w:
        if(i==j):
            count = count+1;
    s1[i] = count
    count = 0

print(s1)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 1, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

In [69]:
```
d1 = {1: 'A', 2: 'B', 3: 'C'}
d2 = {}
for i,j in d1.items():
    d2[j] = i
print(d2)
```

```
{'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

In [89]:

```
d1 = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
s1 = 0
s2 = []
for i in d1:
     sum1 = d1.get(i)
     for j in sum1:
            s1 += j
     s2.append(s1)
     s1 = 0
m = max(s2)
for i in d1:
     sum1 = d1.get(i)
     for j in sum1:
            s1 += j
     if(s1==m):
         print(i)
```

```
Amit
```

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

In [94]:

```
d1 = {1: 11, 2: 20, 3: 15, 4: 40}
s1 = []
for i in d1:
    if(d1.get(i)%2==0):
        s1.append(i)
print(s1)
```

```
[2, 4]
```

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

In [95]:

```
d1 = {1: 'A', 2: 'B', 3: 'C'}
d2 = {2: 'X', 3: 'Y', 4: 'Z'}
s2 = []
for i in d1:
    for j in d2:
        if(i==j):
            s2.append(i)
print(s2)
```

```
[2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

```
Key Not Found
```

In [103…

```
d1 = {'a': 5, 'c': 8, 'e': 2}
key = input("enter key :")
count=0

for i in d1:
    if(i==key):
```

```
        count +=1
        print(d1.get(i))
if(count==0):
        print("not found")
```

8

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

In [106…

```
s = ['Amit', 'Riya', 'Neha', 'Karan']
m = [78, 72, 85, 60]
s2 = []
d2 = {}
for i in m:
    if(i>75):
        s2.append(i)
d2 = dict(zip(s,s2))
d2
```

Out[106…

```
{'Amit': 78, 'Riya': 85}
```

In [107…

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]

result = {students[i]: marks[i] for i in range(len(students)) if marks[i] >= 75}

print(result)
```

```
{'Amit': 78, 'Neha': 85}
```

In [ ]: